Making Effort Observable: AEIC-Mediated Mechanism Design for Multi-Agent LLM Deliberation

Tiancheng Ma[1]

School of Computer Science
author@university.edu

Abstract. Multi-agent language-model systems usually optimize the final aggregate while leaving the behavior that produced it unobserved. This is problematic when a complete, independently reasoned report is more costly than a short answer or imitation. We introduce a *game-driven observable semantic-consensus mechanism* that uses Assumption–Evidence–Inference–Conclusion (AEIC) reports as a parseable interface between a mechanism designer and LLM agents. Before solving, each agent observes a public credit–cost rule and chooses among low-effort reporting, structured reasoning, verification, and peer imitation. The chosen action controls the actual generation procedure and is logged before settlement, enabling a direct follower-response test rather than retrospective response scoring. AEIC is treated as an observation and settlement interface, not assumed to be an intrinsically accuracy-improving prompt template.

Across 450 paired test questions from StrategyQA, MMLU-Hard, and TruthfulQA, a static process contract over the AEIC interface increases high-effort choices from 0.46% to 56.93%; under the joint contract, execution-prompt, and tiered-resource configuration, majority-vote accuracy rises from 68.00% to 82.22% (+14.22 percentage points; Holm-adjusted McNemar $p = 1.23 \times 10^{-11}$). An outcome-only reward raises effort only to 3.08% and does not improve accuracy. A matched-compute control does not reproduce the contract's behavioral response; in a separate interface test, AEIC outperforms free text but not generic structure. Reward and cost sweeps produce the predicted effort response, field ablations do not establish the necessity of any single AEIC field, and a held-out GLM backend reproduces neither gain. Eight frozen suites plus two supplementary experiments contain 6,360 formal question-level records. The supported claim is bounded: public process contracts can change follower resource actions, while the observed quality gain belongs to the joint contract-and-execution configuration rather than to AEIC formatting alone.

Keywords: Multi-agent LLM systems · Mechanism design · Structured reasoning · Deliberation · Follower response

1 Introduction

LLM agents can debate, critique, vote, and synthesize one another's outputs. Yet the aggregate is only as useful as the behavior that produced it. A short answer, an independently reasoned report, a verification pass, and imitation of a visible peer have different costs and informational value. Most multi-agent evaluations observe only the final answer, making these production choices invisible and leaving the system designer unable to distinguish genuine deliberation from cheap agreement.

This is a mechanism-design problem before it is an aggregation problem. A designer must announce what behavior will receive credit before the agents act; the agents must then choose whether the expected credit justifies the effort. Retrospectively weighting an already-generated response cannot establish this follower response, because the generation policy never observed the rule. Similarly, changing communication topology controls where information flows but not whether agents invest in producing independently checkable information.

We introduce a **game-driven observable semantic-consensus mechanism** and use Assumption, Evidence, Inference, and Conclusion (AEIC) reports as its parseable observation interface. Before solving, every follower observes a public offer specifying the credit rule, action costs, information policy, and verification budget. It then declares one of four actions: low effort, structured reasoning, verification, or imitation. The declared action selects a real generation path; the resulting report, resource use, credit, cost, and reputation update are logged. This ordering turns the broad Stackelberg intuition of commitment followed by response into a falsifiable experiment. AEIC makes process signals parseable and contractible; the treatment is the ex-ante contract and its binding to execution resources.

Our confirmatory comparison pairs 450 held-out questions across StrategyQA, MMLU-Hard, and TruthfulQA under three rules. Static AEIC increases high-effort choices from 0.46% to 56.93%, while accuracy under the joint execution configuration rises from 68.00% to 82.22%. Outcome-only reward raises effort only to 3.08% and leaves accuracy statistically unchanged. The behavioral comparison supports a response to a locally legible process rule; because the conditions also alter execution prompts, token budgets, and backend routing, the quality gain cannot yet be attributed to AEIC formatting or credit language alone.

We stress-test that interpretation instead of relying on one headline comparison. Reward and cost sweeps test directional response; evidence, social-only, copy-pressure, and counterevidence environments vary what agents can observe; signal and history ablations probe the implemented contract; two additional seeds and three-to-five-agent panels test stability. Finally, a held-out GLM backend does not respond to the same contract. This negative result prevents a universal claim and points to backend-specific contract calibration as the central unsolved problem.

Our contributions are:

1. an executable leader–follower protocol in which public rules precede logged action choices and action choices control real resource paths;
2. paired evidence that a process contract expressed over the AEIC interface changes follower resource actions and, under the joint execution configuration, accompanies improved collective accuracy beyond an outcome-only reward;
3. a fully frozen eight-suite evaluation, including dose response, information controls, ablations, robustness, and an explicit non-transfer result that defines the claim boundary.

2 Related Work

2.1 Sampling, Voting, and Multi-Agent Deliberation

Early test-time scaling methods aggregate repeated samples. Self-consistency votes over independently sampled reasoning paths [16], while electoral aggregation seeks to preserve candidate diversity under alternative voting rules [24]. These methods reduce single-sample variance, but normally treat every sample as an already-produced information object and do not distinguish the cost of a short answer, independent evidence production, or additional verification.

Multi-agent debate replaces independent sampling with interaction. Du et al. let model instances propose, exchange arguments, and revise answers over multiple rounds [6]; FORD studies inter-model consistency and domination by stronger agents [20]; ChatEval, AgentVerse, and role-based systems organize debaters, critics, and evaluators into task workflows [2,3]. Later work varies personas, reflection, debate modes, voting, and dynamic consensus [23,4,11]. These studies primarily ask how produced outputs should communicate and aggregate. We move the decision boundary earlier and ask how an announced rule changes the action and resource path that produces a report.

2.2 Conformity, Lost Dissent, and Communication Risk

Interaction is not uniformly beneficial. Homogeneous model panels exhibit sycophancy, conformity, and premature consensus, allowing correct minority views to disappear [12,15]; consensus and diversity show task-dependent trade-offs [19]. Adversarial collaboration further demonstrates that a persuasive erroneous agent can contaminate peers through debate [1]. Communication rounds, topology, or majority agreement therefore do not by themselves establish informative deliberation. A system must also record who observed which reports, whether a stance change brought new evidence, and whether alternatives received comparable review.

Our confirmatory design holds the synchronous panel and information flow fixed while changing only the public pre-action rule. Controlled suites then expose new evidence, unsupported majority stances, copy pressure, or counterevidence. This complements failure-mode studies: prior work characterizes how errors propagate, whereas we test whether an observable process contract changes information production before propagation.

2.3 Structured Reasoning, Verification, and Observable Reports

Chain-of-Thought externalizes intermediate steps, Tree-of-Thoughts organizes search, and ReAct interleaves reasoning with tool actions [18,21,22]. They demonstrate the value of decomposed computation and tool grounding, but structured prose is not automatically factual and does not establish independent verification. Free-form reviews additionally suffer from missing fields, unnormalized conclusions, and untraceable evidence.

AEIC is an audit interface rather than a request for private chain-of-thought. It stores concise public assumptions, evidence, inference, and conclusion fields and converts completeness, legal stance, evidence increment, verified support, useful correction, and unsupported reversal into settlement signals. We separately compare free text, a generic analysis–answer structure, and AEIC under matched budgets so that benefits of structure in general are not attributed to the AEIC labels.

2.4 Mechanism Design and Principal–Agent Views of LLMs

Mechanism design studies rules committed before participants act. Existing work formulates auctions for LLM-mediated generated content [7], uses statistical contracts to incentivize measurable text quality [14], and frames multi-agent AI through principal–agent information asymmetry [13,10]. Stackelberg incentives are also used in federated learning, crowdsourcing, and resource allocation [8,9,17].

Unlike retrospective scoring, our contract is visible to followers before solving, and the declared action selects different prompts, token caps, call counts, and verification paths. The empirical object is the induced action distribution of a black-box model under a fixed public contract, not an assumed human-like utility function. We use Stackelberg timing to organize a falsifiable experiment but do not claim a closed-form equilibrium, general incentive compatibility, or a globally optimal contract.

2.5 Positioning

Prior multi-agent work richly studies roles, topology, debate rounds, voting, and judges; structured reasoning provides intermediate representations; mechanism design provides commitment and response. The missing executable connection is that the rule must be visible before generation, the action must control a real resource path, the natural-language report must become a parseable settlement object, and retries and information exposure must remain traceable. We use AEIC as the reporting interface, a public contract plus action router as the treatment, and paired follower response as the primary evidence. The contribution therefore does not rely on assuming that debate always beats voting or that structured formatting is intrinsically accuracy improving.

3 AEIC as an Observable Report Space

For question x, agent i submits

$$r_i = (A_i, E_i, I_i, C_i), \tag{1}$$

where A_i states the operative assumptions, E_i is a list of specific evidence items, I_i connects evidence to the answer, and C_i contains an explicit final stance. The report is serialized as a strict JSON object with exactly these four fields. This design supplies three observables that free-form deliberation usually conflates: whether the report is structurally complete, what evidence is exposed to later agents, and what answer is finally adopted.

The fields have distinct audit roles. A_i exposes premises so that disagreement can be localized to facts, definitions, or task interpretation. E_i names support from the question, authorized history, or verification calls. I_i makes the transition from evidence to stance inspectable, and C_i is restricted to the task label space for deterministic aggregation. We store concise audit reports rather than requesting or using private chain-of-thought.

AEIC is not treated as proof of correctness. A complete report may be coherently wrong, and an agent may imitate a peer while filling every field. Instead, AEIC provides a locally checkable surface on which a mechanism can condition credit. For a report r, define completeness

$$q_{\text{comp}}(r) = \frac{1}{4} \sum_{z \in \{A,E,I,C\}} \mathbf{1}[z \text{ is non-empty}], \tag{2}$$

and let $q_C(r)$ indicate whether C yields a valid task stance. The implemented static rule uses only these process-level signals. It never receives the ground-truth answer as an input to the follower's choice prompt.

The parser additionally returns

$$s(r_i, h_i) = \big(q_{\text{comp}}, q_C, q_{\text{cons}}, q_{\text{nov}}, q_{\text{ver}}, q_{\text{corr}}, q_{\text{flip}}\big), \tag{3}$$

covering completeness, legal conclusion, within-report consistency, evidence increments, verifiable support, useful correction, and unsupported reversal. Novelty is a conservative token increment rather than factual novelty. Verified evidence must resolve to the question, a deterministic tool result, or another traceable record. Syntactic, semantic, and task validators are separated; answer keys are available only to post-generation outcome settlement. A repair call may restore keys and types, but may neither change the attempted stance nor introduce facts, and both raw and repaired outputs remain in the trace.

The distinction between *observable process* and *truth* is deliberate. Outcome-only credit requires the environment to know correctness after the task, whereas process credit can be announced and checked from the report itself. The comparison between these two rules asks whether a reward attached to a delayed outcome has the same behavioral effect as one attached to an immediately legible production process.

4 AEIC-Mediated Leader–Follower Mechanism

4.1 Timing and Action Space

The mechanism designer is a leader that commits to a public offer m before generation. A follower observes the offer and chooses

$$a_i \in \mathcal{A} = \{\textsc{Low}, \text{AEIC}, \textsc{Verify}, \textsc{Imitate}\}. \tag{4}$$

The four actions respectively trigger a short low-cost report, independent complete AEIC reasoning, a draft followed by an additional verification call, or a short synthesis of visible peer information. Thus, action labels are not retrospective annotations: they select different prompts, token limits, and in the tiered executor potentially different model resources.

An offer contains a visible credit rule g_m, action costs $c_m(a)$, an information policy, and a verification budget. The choice prompt explicitly asks the follower to maximize expected net utility,

$$U_i(a_i; m, h_i) = \mathbb{E}[g_m(r_i, y, h_i) \mid a_i, m, h_i] - c_m(a_i), \tag{5}$$

where h_i denotes visible history and y is available to the settlement environment only when the rule requires outcome verification. We make no claim that the language model computes Eq. (5) exactly; the observable implication is whether its action distribution changes when the revealed reward or cost changes.

We write the contract as $m = (g_m, c_m, P_m, B_m)$, where P_m is the cross-round visibility policy and B_m is the verification budget. The contract is inserted before problem solving and held fixed for the question. A deterministic mapping $\phi(a)$ translates a declaration into a prompt template, token cap, call count, peer-history permission, and verification branch. Actions are therefore resource controls rather than labels assigned after generation.

4.2 Compared Rules

The confirmatory experiment compares three offers while holding questions, seeds, panel size, number of rounds, and information policy fixed.

No mechanism. No explicit credit is offered. The action costs are visible.

Outcome only. Credit is paid for a correct final stance. The report's internal structure does not affect settlement.

Static AEIC. Credit is paid for AEIC completeness and a valid explicit conclusion. The rule is fixed across agents and rounds and does not inspect the answer key.

Let $H(a) = \mathbf{1}[a \in \{\text{AEIC}, \textsc{Verify}\}]$. Our primary behavioral estimand is the paired change in $\mathbb{E}[H]$ under a revealed rule. The primary decision estimand is the paired change in question-level majority-vote correctness. Token use is a resource outcome, not a substitute for the declared action.

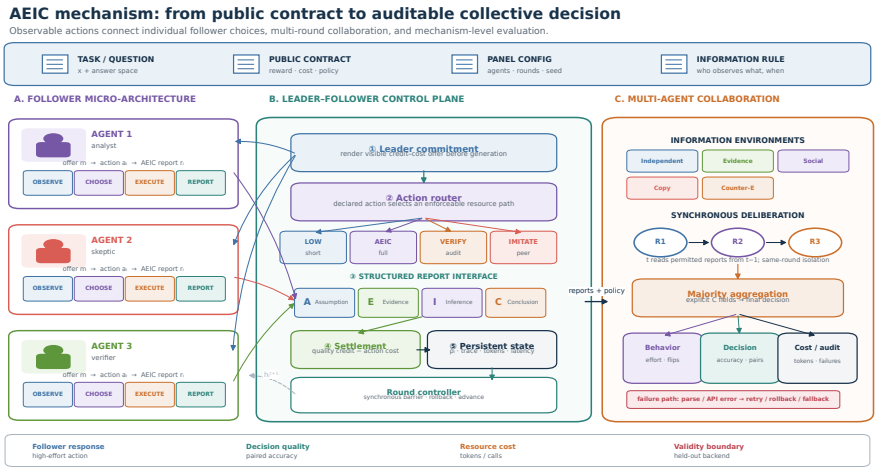


Fig. 1. The AEIC mechanism architecture. A public contract is rendered before followers choose resource-bearing actions. The selected path produces a structured AEIC report, after which the controller settles credit and cost, records state, and exposes only policy-permitted information to later rounds. The right-hand audit separates follower behavior, collective decision quality, and resource use.

4.3 Settlement and Longitudinal State

After generation, the controller computes credit, subtracts the announced action cost, and updates reputation by

$$\text{credit}_i^t = \sum_k w_k s_k(r_i^t, h_i^t, y), \qquad \text{utility}_i^t = \text{credit}_i^t - c_m(a_i^t). \tag{6}$$

Process contracts do not read y; outcome contracts access it only after generation. Weights, costs, and budgets are frozen in the run manifest rather than adjusted from the current answer.

$$\rho_i^t = \delta \rho_i^{t-1} + g_m(r_i^t, y, h_i^t) - c_m(a_i^t), \qquad \delta = 0.9. \tag{7}$$

The static rule does not adapt its coefficients to reputation, but the complete state is retained to support history ablations and future adaptive contracts. Ground truth is withheld from settlement until the final round. Failed attempts are rolled back; retries and repair calls remain in the action trace.

The collective stance is formed only from legal C fields. Confirmatory experiments retain a third-round simple majority so that a new aggregator is not confounded with a new production incentive. Ties, invalid conclusions, and terminal failures follow frozen deterministic policies.

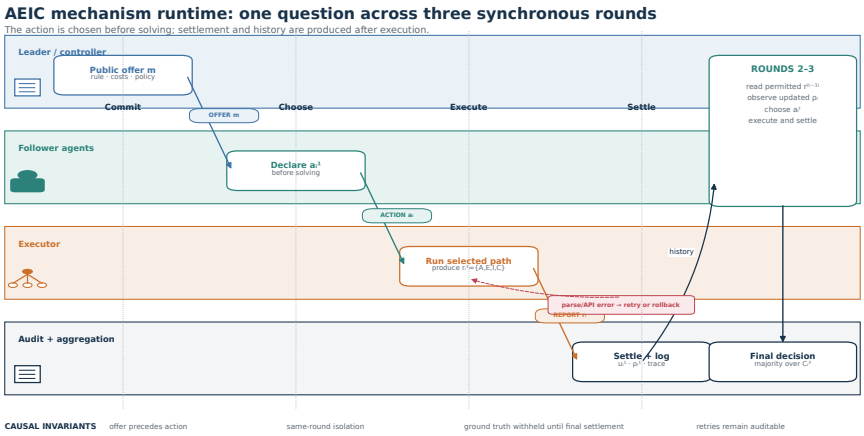


Fig. 2. Runtime sequence for one question. The four horizontal lanes separate controller commitment, follower choice, enforced execution, and settlement. Cross-round history is generated only after the current round has been logged, preserving same-round isolation.

5 Experimental Protocol and Audit Trail

Synchronous panels. Each question is processed by five agents for three synchronous rounds. In round t, an agent can observe only the permitted subset of round-$t-1$ reports; agents in the same round cannot observe one another. The confirmatory suite uses the independent information condition, so no peer content is visible. Controlled information suites later expose one new evidence item, only a majority stance, full peer reports with copy pressure, or stance-opposing evidence.

Figure 2 makes the implemented ordering explicit. The controller commits first, each follower declares an action before solving, the executor runs the corresponding path, and only then does settlement create the history visible in subsequent rounds. Parse or API failures follow an auditable retry or rollback branch rather than silently changing the protocol.

Choice before execution. Every agent first receives only the public contract, its visible reputation, the permitted information, and the question. It returns a single action. A second call executes the selected action. VERIFY adds a third audit-and-revision call. A formatting repair may be called only when JSON parsing fails; it is instructed to preserve the attempted conclusion and add no facts. All calls and token counts remain logged.

Pairing and aggregation. All mechanisms for a question share the question-level seed. The final panel decision is majority vote over explicit C fields in the third round. High effort is defined before analysis as AEIC or VERIFY.

Statistical tests operate at the natural paired unit: agent-round for the binary action and question for final correctness. We use exact McNemar tests for paired binary outcomes and paired rank tests for continuous outcomes, applying Holm correction within each registered contrast family.

Frozen provenance. Records use a versioned schema that requires the offer, chosen action, report, visible peers, reward, realized cost, reputation snapshots, raw response, action trace, latency, and token use before a record can be marked complete. Each suite passes an expected-versus-present matrix check, is frozen with SHA-256 provenance, and is independently recomputed from canonical question-level JSONL. The final manifest contains eight complete suites and 5,850 question-level records; no suite has missing, incomplete, unexpected, route-invalid, or schema-invalid records.

Failure detection and resumption. Every unit is keyed by condition, dataset, item, and seed. On restart, the collector validates configuration hashes, schemas, and the expected item set, then reruns only missing or explicitly failed units. Duplicate keys, cross-condition input mismatch, answer leakage into prompts, missing terminal responses, and anomalous token accounting fail validation. API timeouts use bounded backoff; exhausted units preserve their error status rather than masquerading as successes. Analysis reads only validated frozen JSONL and checks every denominator.

Information isolation. Adapters construct the question and authorized history, while answer keys are passed only to settlement and offline evaluation. Choice, generation, and repair calls never receive the key. Agents in a round are committed as a batch; history and reputation are updated only after the full round, preventing order-dependent same-round leakage.

6 Identification and Claim Boundaries

The AEIC mechanism uses a Stackelberg *ordering*, not a claimed closed-form equilibrium: the designer commits to m, then followers respond. Because the LLM policy is a black box, the empirical object is the induced action distribution $\pi(a \mid x, m, h)$.

Follower-response contrast. For paired task instances, changing m while fixing the question, seed, panel configuration, rounds, and information policy identifies the intention-to-treat effect of the announced rule on the implemented model policy. A positive difference in high-effort actions establishes rule responsiveness; it does not by itself prove utility maximization or incentive compatibility for arbitrary agents.

Outcome-only control. The contrast between Static AEIC and Outcome Only separates process observability from the mere presence of reward language. If both rules offered incentives but only the process rule changed effort and accuracy, the evidence favors the locally checkable report interface as the operative component. It does not isolate a unique contribution for each of the four AEIC fields; the registered signal ablations are descriptive bounds on the implemented contract.

Dose response. For a fixed backend and domain, Eq. (5) predicts that increasing visible reward should weakly increase high-effort choices and increasing visible effort cost should weakly decrease them. We test these directional implications without fitting a structural utility model. The boundary suite combines frozen low/high settings with a same-domain development center point, so it supports monotonicity rather than an unbiased estimate of the confirmatory treatment effect.

External validity. Prompt-mediated costs and credits have semantics supplied by the backend. Consequently, a contract that changes π for one model family need not transfer to another. We therefore treat the held-out backend as a decisive scope test and report failure to transfer in the main text. We do not claim general equilibrium existence, global leader optimality, monetary incentive compatibility, or topology-independent effectiveness.

Falsifiable predictions. For a responsive follower, increasing visible process credit should not reduce $\Pr(H(a) = 1)$, increasing the cost of the same action should not increase it, and an outcome-only promise should not reproduce the same response if locally satisfiable production rules are operative. Reward scans, cost scans, and the three-rule confirmatory comparison test these implications. Their failure would weaken the mechanism account rather than being relabeled as a prompting success.

What is and is not identified. The confirmatory treatment jointly changes public credit language, the action-choice prompt, and action-conditioned execution. It identifies the total effect of this executable contract, not a field-level cause of accuracy. The compute-matched control probes the "more tokens only" explanation, while the report-interface study compares AEIC with free text and a generic structure. Together they narrow the account without establishing an independent causal effect for every AEIC field.

7 Experiments

7.1 Setup

We use 150 held-out questions from each of StrategyQA, MMLU-Hard, and TruthfulQA. Each condition uses five agents, three synchronous rounds, and DeepSeek-V3 as the primary backend [5]. The confirmatory suite contains $450 \times$

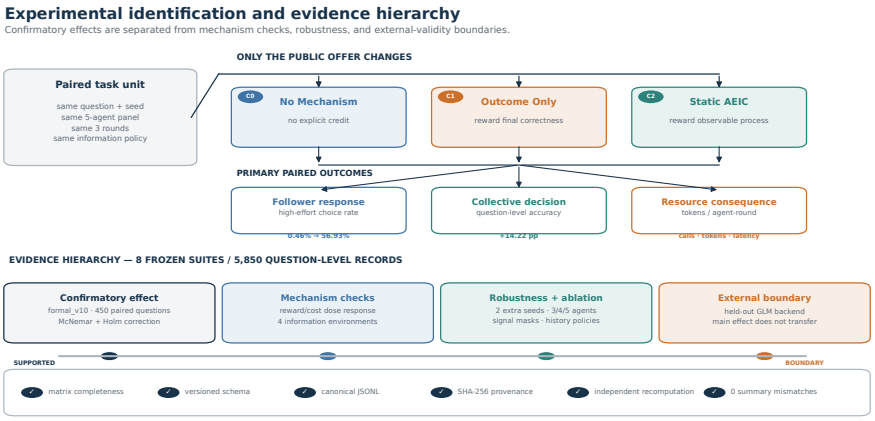


Fig. 3. Experimental identification and evidence hierarchy. Paired conditions isolate the public rule; primary outcomes measure follower response, collective accuracy, and resource use. Eight frozen suites cover confirmation, mechanism checks, robustness, ablation, and the explicit non-transfer boundary.

3 = 1,350 question-condition records and 20,250 agent-rounds. Questions, seeds, panels, rounds, and aggregation are paired across No Mechanism, Outcome Only, and Static AEIC; only the visible offer changes. Final decisions use majority vote over the explicit third-round conclusions.

The two primary outcomes are high-effort choice rate—the share of AEIC and VERIFY actions—and question-level decision accuracy. We additionally report tokens per agent-round, parsing and collection integrity, and domain-level effects. Exact McNemar tests are used for paired binary outcomes and Holm-adjusted within the registered comparison family.

Figure 3 summarizes the identification logic and the status of the evidence. The confirmatory contrast changes only the public offer within paired task units. Mechanism checks and robustness suites support interpretation of that contrast, whereas the held-out backend is treated as an external-validity boundary rather than folded into a pooled positive claim.

7.2 Confirmatory Follower-Response Test

Static AEIC changes both follower behavior and the collective decision (Table 1; Figure 4). High-effort choice increases by 56.47 percentage points. Accuracy rises by 14.22 points, with 78 questions corrected and 14 harmed relative to No Mechanism (Holm-adjusted McNemar $p = 1.23 \times 10^{-11}$). The additional 145.5 tokens per agent-round are consistent with the intended resource cost of more complete work.

Outcome Only is an informative control. It raises high effort by only 2.62 points and accuracy changes by −0.67 points (27 questions corrected, 24 harmed;

Table 1. Confirmatory paired test across 450 questions. High effort is measured over 6,750 agent-rounds per rule; accuracy is measured over 450 questions.

Rule	High effort	Accuracy	Tokens/agent-round
No mechanism	0.46%	68.00%	663.3
Outcome only	3.08%	67.33%	689.2
Static AEIC	**56.93%**	**82.22%**	808.8

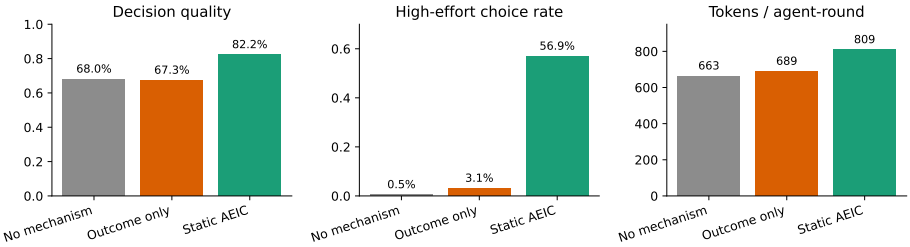


Fig. 4. Formal paired comparison. Static AEIC changes the declared effort action and improves final accuracy; an outcome-only reward does not improve accuracy.

$p = 0.780$). Therefore, the main effect cannot be summarized as "telling the model that a reward exists." What differs is whether the rule exposes a criterion the follower can satisfy through the chosen production process.

The gain occurs in all three domains but is heterogeneous: +4.0 points on StrategyQA, +20.0 on MMLU-Hard, and +18.7 on TruthfulQA. The corresponding increases in high effort are 43.5, 73.7, and 52.2 points. This heterogeneity argues against interpreting the pooled effect as a task-independent constant.

At the paired-question level, Static AEIC produces 78 wrong-to-correct and 14 correct-to-wrong transitions, for 64 net corrections. It adds 2,182 tokens and 0.80 calls per question on average, or 15,343 tokens and 5.64 calls per net correction. These quantities make the resource cost of the quality gain explicit rather than presenting it as a free improvement.

7.3 Report Interface and Mechanism Decomposition

To separate the AEIC interface from contract execution, we hold model, call count, and maximum token budget fixed on 90 paired questions and compare free text, a generic analysis–answer structure, and AEIC. Their accuracies are 72.22%, 82.22%, and 85.56%, respectively. AEIC exceeds free text by 13.33 points (Holm $p = 0.00836$), but exceeds generic structure by only 3.33 points ($p = 0.453$). The supported role of AEIC is therefore a common, parseable, contractible semantic interface; the current experiment does not show an accuracy advantage independent of generic structuring.

We then fix the DeepSeek backend on 60 paired questions and compare no-contract fixed resources, contract fixed resources, no-contract matched com-

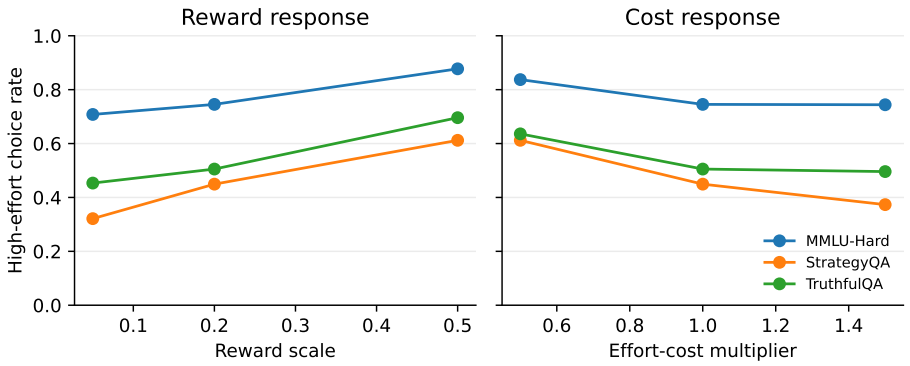


Fig. 5. Directional response to the public offer. Within every domain, effort rises with revealed reward and falls with action cost.

pute, and the full mechanism. High-effort rates are 14.44%, 81.11%, 17.78%, and 81.11%; accuracies are 85.00%, 88.33%, 86.67%, and 91.67%. Relative to matched compute, the full mechanism raises high effort by 63.33 points ($p = 2.82 \times 10^{-33}$), while actual token use is not significantly different ($p = 0.229$). Thus the behavioral response is not explained by mechanically adding compute. The accuracy difference is +5 points, but with only three wrong-to-correct and zero correct-to-wrong transitions ($p = 0.25$), it remains directional component evidence.

7.4 Reward–Cost Response

For every domain, high-effort choice is monotone non-decreasing as the revealed reward scale moves from 0.05 to 0.2 to 0.5, and monotone non-increasing as the effort-cost multiplier moves from 0.5 to 1.0 to 1.5 (Figure 5). At the pooled frozen extremes, high effort rises from 49.42% at reward 0.05 to 72.84% at reward 0.5; it falls from 69.51% at cost multiplier 0.5 to 53.78% at 1.5. Because the middle points are same-domain development observations whereas the extremes are separately frozen, this is a directional mechanism check rather than a second confirmatory effect estimate.

7.5 Controlled Information Environments

We next vary only what peer information is available after the first round: one new evidence item, a majority stance without evidence, full reports plus copy pressure, or stance-opposing evidence. Across these four environments, Static AEIC raises pooled high effort from 0.86% to 49.41%. Condition-specific rates under Static AEIC range from 40.89% (social only) to 59.33% (copy pressure), compared with 0.44%–1.47% without a mechanism (Figure 6). These manipulations confirm that the report interface remains behaviorally operative under distinct information structures. They do not establish that every information intervention improves accuracy or eliminates imitation.

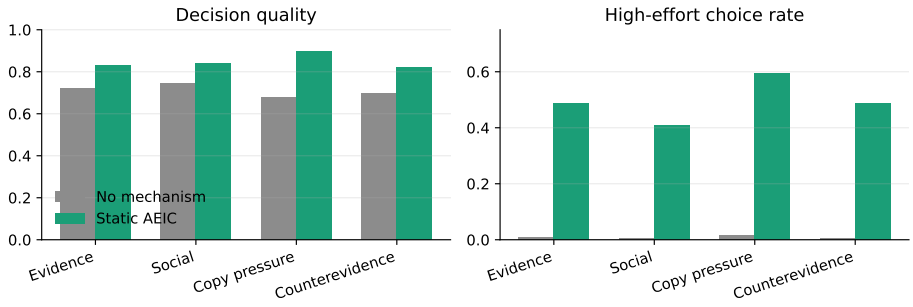


Fig. 6. High-effort actions under controlled information environments. Static AEIC shifts behavior in all four settings.

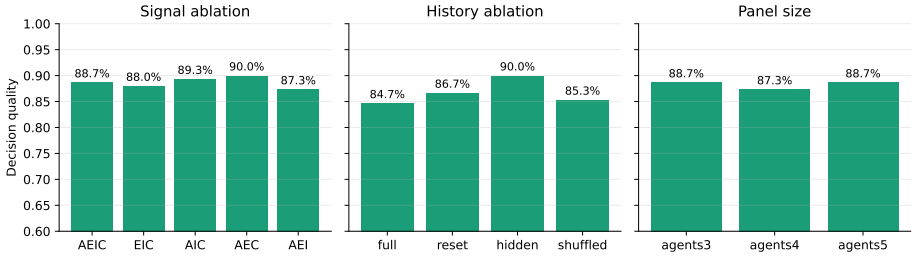


Fig. 7. Signal, history, seed, and panel-size robustness. Component variants are descriptive boundaries rather than isolated necessity proofs.

7.6 Robustness, Ablation, and External Validity

Two additional seeds reproduce the main direction: Static AEIC reaches 85.67% accuracy versus 71.00% without a mechanism (+14.67 points; Holm-adjusted $p = 2.40 \times 10^{-9}$). With three, four, and five agents, its accuracy lies between 87.33% and 88.67%. Signal-mask and history-policy variants are summarized in Figure 7; they bound the current implementation but do not show that each AEIC field is individually necessary. In particular, removing a field can change how the backend interprets the entire visible contract, so these are intervention ablations rather than modular proofs.

The held-out GLM backend is the strongest boundary result. Static AEIC barely changes high effort (0.58% to 0.93%, $p = 0.134$) and yields 69.33% accuracy versus 70.67% without a mechanism (Holm-adjusted $p = 0.791$). The same words and numerical schedule therefore do not define the same effective contract across model families. This failure is reported in the main text because it changes the scope of the contribution: the AEIC mechanism is an experimentally validated mechanism for the tested primary backend, not a universally calibrated incentive language.

7.7 Integrity and Reproducibility

Across the eight frozen suites, all 5,850 expected question-level records are present. The report-interface and mechanism-decomposition supplements add 270 and 240 formal records, respectively, for 6,360 records in total. Every frozen suite passes schema and route validation and has an independently recomputed bundle with no summary mismatch; all 240 decomposition records also pass integrity checks. In the confirmatory data, API terminal failure is zero and cost logging is complete; rare parse failures are retried and remain visible in attempt logs. The manifest stores canonical hashes for each suite, preventing silent replacement of the analyzed records.

8 Discussion

Mechanism, not formatting alone. AEIC is useful here because it creates contractible observables, not because four labeled fields guarantee good reasoning. Static AEIC changes behavior when agents see a rule tied to those observables; Outcome Only shows that generic reward language is insufficient. At the same time, the signal ablations do not justify claiming that every field is independently indispensable.

Why the negative transfer matters. The GLM result indicates that prompt-mediated costs and credits are interpreted through a backend-specific policy. The contract's nominal numbers are therefore not yet a common currency. A credible next step is a calibration stage that estimates each backend's response surface on a development set before testing a fixed contract on held-out tasks. Mixed-model panels should be evaluated only after this calibration problem is addressed.

Relation to topology and deliberation. Production incentives and communication topology solve different problems. The former governs whether informative work is produced; the latter governs who can observe it and at what cost. The controlled information experiments begin to connect the two without turning topology into the paper's main causal variable. Future work can optimize information flow conditional on a calibrated reporting mechanism.

Limitations. The primary effect is established for one backend, three binary-answer QA domains, one prompt-mediated action menu, and majority aggregation. Declared actions are behavior generated by an LLM under instructions, not evidence of human-like preferences. Credits are bookkeeping quantities rather than monetary transfers, and process completeness is not factual verification. The confirmatory experiment tests a fixed static rule, not adaptive Stackelberg optimality or a general equilibrium. Finally, additional API calls and longer reports incur real compute cost even when token count is used only as a proxy.

Behavior before accuracy. The primary mechanism question is whether the contract changes follower action; collective correctness is a downstream outcome of the joint contract and execution path. The defensible claim is not that four AEIC labels inherently improve reasoning, but that an observable, enforceable, ex-ante process rule induces different resource behavior on the tested backend and is accompanied by a costly quality improvement.

Cost effectiveness. Static AEIC consumes 145.5 additional tokens per agent-round and yields 64 net question corrections relative to no mechanism. This accounting enables risk-sensitive deployment: low-risk items can retain cheap paths, while disputed or high-risk items receive verification budget. We have not solved the optimal routing threshold, so these figures expose the trade-off rather than establish optimal efficiency.

Broader implications. The framework is most appropriate when intermediate work can be made locally inspectable and when the executor can enforce meaningful differences between actions. It should not be used to manufacture confidence from structurally complete but unverifiable reports. Logging the offer, action, execution path, and settlement is therefore part of the contribution, not merely an engineering detail.

9 Conclusion

This work uses AEIC structured reports as an observable interface for leader–follower experimentation and binds ex-ante public contracts to real execution paths. A public Static-AEIC rule substantially changes the primary backend's resource actions, while collective accuracy improves under the joint contract-and-execution configuration; an outcome-only reward does not yield the same pattern. Controlled information experiments support the behavioral measurement value of AEIC-derived signals, but field ablations establish neither the independent necessity of a field nor an accuracy effect of AEIC formatting itself. Dose response, robustness suites, and frozen provenance support this bounded interpretation, and the held-out-backend failure shows that effective contracts require model-specific calibration. The next identification sequence is report-interface validation, contract/resource decomposition, and only then finite contract selection.

Accordingly, the paper identifies empirical follower response under fixed public contracts rather than proving a general Stackelberg equilibrium. Its reusable artifacts are the report schema, choice-before-execution resource enforcement, process-signal settlement, longitudinal reputation state, and recomputable traces. Future work will estimate reward–cost response surfaces on frozen tasks, define a leader objective over quality, resources, and failure risk, and select only within a finite contract space; every selected contract must be recalibrated on new model backends.

References

1. Amayuelas, A., Yang, X., Antoniades, A., Hua, W., Pan, L., Wang, W.Y.: Multiagent collaboration attack: Investigating adversarial attacks in large language model collaborations via debate. In: Findings of the Association for Computational Linguistics: EMNLP 2024 (2024). https://doi.org/10.18653/v1/2024.findings-emnlp.407
2. Chan, C.M., Chen, W., Su, Y., Yu, J., Xue, W., Zhang, S., Fu, J., Liu, Z.: ChatEval: Towards better LLM-based evaluators through multi-agent debate. In: Proceedings of the 12th International Conference on Learning Representations (ICLR) (2024), https://arxiv.org/abs/2308.07201
3. Chen, W., Su, Y., Zuo, J., Yang, C., Yuan, C., Chan, C.M., Yu, H., Lu, Y., Hung, Y.H., Qian, C., Qin, Y., Cong, X., Xie, R., Liu, Z., Sun, M., Zhou, J.: AgentVerse: Facilitating multi-agent collaboration and exploring emergent behaviors. In: Proceedings of the 12th International Conference on Learning Representations (ICLR) (2024), https://arxiv.org/abs/2308.10848
4. Choi, H.K., Zhu, X., Li, Y.: Debate or vote: Which yields better decisions in multi-agent large language models? CoRR **abs/2508.17536** (2025). https://doi.org/10.48550/arXiv.2508.17536
5. DeepSeek-AI: DeepSeek-V3 technical report. CoRR **abs/2412.19437** (2024). https://doi.org/10.48550/arXiv.2412.19437, https://arxiv.org/abs/2412.19437
6. Du, Y., Li, S., Torralba, A., Tenenbaum, J.B., Mordatch, I.: Improving factuality and reasoning in language models through multiagent debate. In: Proceedings of the 41st International Conference on Machine Learning (ICML) (2024), https://arxiv.org/abs/2305.14325
7. Dütting, P., Mirrokni, V., Leme, R.P., Xu, H., Zuo, S.: Mechanism design for large language models. CoRR **abs/2310.10826** (2023), https://arxiv.org/abs/2310.10826
8. Fu, Y., Dong, M., Zhou, L., Li, C., Yu, F.R.: A distributed incentive mechanism to balance demand and communication overhead for multiple federated learning tasks in IoV. IEEE Internet of Things Journal **11**(4), 5997–6013 (2024)
9. Guo, W., Wang, Y., Jiang, P.: Incentive mechanism design for federated learning with Stackelberg game perspective in the industrial scenario. Computers & Industrial Engineering **186**, 109741 (2023)
10. Kaltenpoth, S., Müller, O.: Getting in contract with large language models – an agency theory perspective on large language model alignment. CoRR **abs/2509.07642** (2025). https://doi.org/10.48550/arXiv.2509.07642, https://arxiv.org/abs/2509.07642
11. Li, Y., Zhu, C., Wang, M., Wu, M., Zhu, M., Hu, X.: Dynamic weighted consensus framework for LLM multi-agent debate. In: Lecture Notes in Computer Science. Springer (2025). https://doi.org/10.1007/978-981-95-3001-4_14
12. Pitre, P., Ramakrishnan, N., Wang, X.: CONSENSAGENT: Towards efficient and effective consensus in multi-agent LLM interactions through sycophancy mitigation. In: Findings of the Association for Computational Linguistics: ACL 2025 (2025). https://doi.org/10.18653/v1/2025.findings-acl.1141
13. Rauba, P., Cepenas, S., van der Schaar, M.: Multi-agent systems should be treated as principal-agent problems. CoRR **abs/2601.23211** (2026). https://doi.org/10.48550/arXiv.2601.23211, https://arxiv.org/abs/2601.23211

14. Saig, E., Einav, O., Talgam-Cohen, I.: Incentivizing quality text generation via statistical contracts. In: Advances in Neural Information Processing Systems 37 (NeurIPS) (2024), https://arxiv.org/abs/2406.11118
15. Swamy, S., et al.: Peacemaker or troublemaker? Sycophancy in multi-agent debate. arXiv preprint (2025)
16. Wang, X., Wei, J., Schuurmans, D., Le, Q.V., Chi, E.H., Narang, S., Chowdhery, A., Zhou, D.: Self-consistency improves chain of thought reasoning in language models. In: Proceedings of the 11th International Conference on Learning Representations (ICLR) (2023), https://arxiv.org/abs/2203.11171
17. Wang, Y., Zhou, Y., Huang, P.Q.: A novel incentive mechanism for federated learning over wireless communications. IEEE Transactions on Artificial Intelligence **5**(11), 5549–5563 (2024)
18. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E.H., Le, Q.V., Zhou, D.: Chain-of-thought prompting elicits reasoning in large language models. Advances in Neural Information Processing Systems **35** (2022)
19. Wu, Z., Ito, T.: The hidden strength of disagreement: Unraveling the consensus-diversity tradeoff in adaptive multi-agent systems. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 15277–15297 (2025). https://doi.org/10.18653/v1/2025.emnlp-main.772
20. Xiong, K., Ding, X., Cao, Y., Liu, T., Qin, B.: Examining inter-consistency of large language models collaboration: An in-depth analysis via debate. In: Findings of the Association for Computational Linguistics: EMNLP 2023. pp. 7572–7590 (2023). https://doi.org/10.18653/v1/2023.findings-emnlp.508
21. Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T., Cao, Y., Narasimhan, K.: Tree of thoughts: Deliberate problem solving with large language models. Advances in Neural Information Processing Systems **36** (2024)
22. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y.: React: Synergizing reasoning and acting in language models. International Conference on Learning Representations (2023)
23. Zhang, J., Xu, X., Zhang, N., Liu, R., Hooi, B., Deng, S.: Exploring collaboration mechanisms for LLM agents: A social psychology view. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics. pp. 14544–14607 (2024). https://doi.org/10.18653/v1/2024.acl-long.782
24. Zhao, X., Wang, K., Peng, W.: An electoral approach to diversify LLM-based multi-agent collective decision-making. In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 2777–2795 (2024). https://doi.org/10.18653/v1/2024.emnlp-main.158